Sentinel SDK — Integration Guide

Instrument request pipelines with anomaly detection in under ten minutes.

1. Installation

Sentinel ships as a single package with zero runtime dependencies. Install from the package index and verify the CLI is on your path:

```
pip install sentinel-sdk
sentinel --version
# sentinel 3.2.1 (python 3.11)
```

2. Quickstart

The minimal integration wraps your handler with a detector. Thresholds adapt automatically after the warmup window:

```
from sentinel import Detector, Policy

detector = Detector( policy=Policy.adaptive(warmup=500), channels=["latency", "error_rate",
"payload_entropy"], )

@detector.guard def handle(request): return downstream.process(request)
```

Performance note: The guard decorator adds ~40μs of overhead per call. For hot paths above 50K RPS, use the batched API described in Section 4.

3. Configuration Reference

All options can be set via constructor, environment, or config file. Environment variables take precedence.

Parameter	Type	Default	Description
`policy`	`Policy`	`adaptive(500)`	Detection policy and warmup window
`channels`	`list[str]`	`["latency"]`	Signals monitored per request
`sink`	`Sink`	`stdout`	Where verdicts are emitted
`sample_rate`	`float`	`1.0`	Fraction of traffic inspected

fail_open	bool	True	Pass traffic if detector errors

Table 2: Constructor parameters.

4. Batched Verdicts

For high-throughput services, evaluate verdicts asynchronously in batches. The detector coalesces up to `max_batch` requests per tick:

```
async def pump(queue, detector): while True: batch = await queue.take(max_batch=256, timeout_ms=5) verdicts = detector.evaluate(batch) for v in verdicts.flagged(): await quarantine.put(v.request_id)
```

Warning: Never block the request path on sink I/O. Configure sink=Sink.buffered(...) in production deployments.

5. Verdict Lifecycle

1. Capture — signals sampled from the live request.
2. Score — policy computes an anomaly score per channel.
3. Decide — scores fold into a verdict with a confidence bound.
4. Emit — verdicts stream to the configured sink.